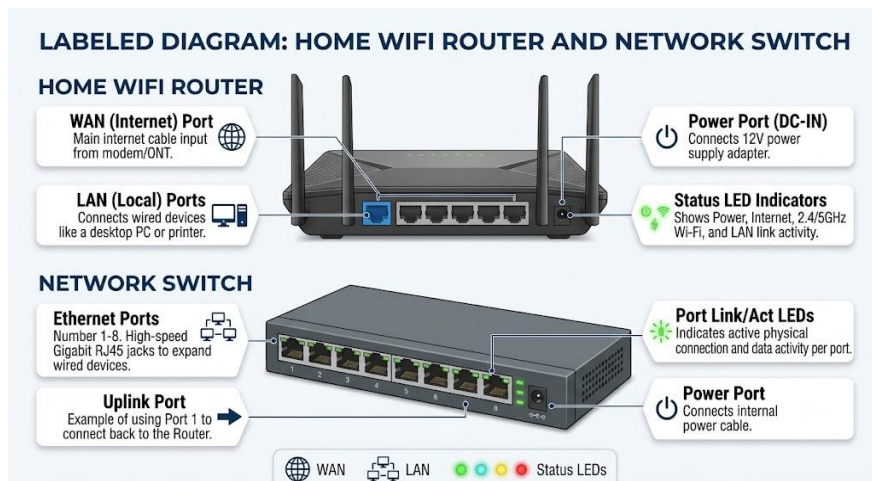


Routers And Switch Theory

Task 1: Labelled Diagram of a Home Wi-Fi Router & Network Switch



- **Home Wi-Fi Router:**
 - **WAN (Internet) Port:** Usually coloured blue or yellow; connects directly to your modem or ISP incoming cable.
 - **LAN Ports (Ports 1–4):** Connects local wired devices such as your PC, TV, or switch using RJ45 Ethernet cables.
 - **Power Port (DC-IN):** Barrel jack where the 12V power supply adapter plugs in.
 - **Status LEDs (Front Panel):** Front indicator lights showing **Power** (device on/off), **Internet/WAN** (online status), **Wi-Fi 2.4G/5G** (wireless activity), and **LAN 1–4** (active wired links).
- **Network Switch:**
 - **Ethernet Ports (RJ45):** Multi-port bank (typically 8, 16, or 24 ports) to plug in local wired devices.
 - **Console / Uplink Port:** Used for management setup (console) or connecting an uplink cable back to the main router.
 - **Power Port:** Where the power cable connects.
 - **Port Link/Activity LEDs:** Small LED above or beside each individual RJ45 port that lights green/amber when connected and flashes during data transfer.

Task 2: 3 Key Differences Between Routers and Switches

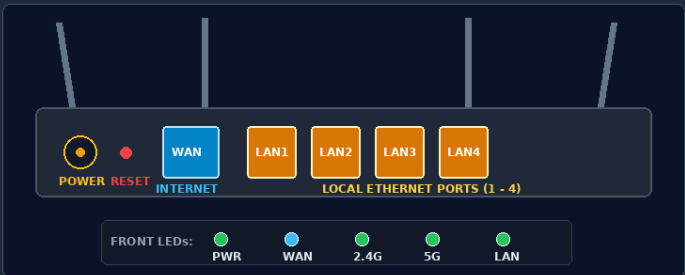
Difference	Router	Switch
1. Primary Function & Network Type	Connects different networks together (links your private Local Area Network / LAN to the public Internet / WAN).	Connects multiple devices together within the same local network (LAN).
2. Addressing & How Traffic Is Handled	Reads IP addresses (Layer 3 - Network Layer) and routes data packets between networks using routing tables.	Reads hardware MAC addresses (Layer 2 - Data Link Layer) to forward data frames directly to specific ports.
3. Traffic Scope & Broadcasts	Limits and controls broadcast domains; directs external and internal traffic through Network Address Translation (NAT) and firewall rules.	Forwards data strictly within one local broadcast domain; does not assign public IPs or translate external internet addresses.

Task 3: Identifying Ports & LEDs on Real/Virtual Images

Task 3: Hardware Port & LED Identification

Detailed annotation of Power Ports, Ethernet Ports, and LED Indicators on a Router and Switch

1. HOME WI-FI ROUTER (Rear & Front Panel)



1. Power Port (DC-IN) [Position: Rear Left Panel]

Barrel jack (12V DC) that connects to the external wall adapter. It energizes the router's internal CPU, memory, and Wi-Fi radios.

2. Ethernet Port #1: WAN (Internet) [Position: Rear Panel (Blue Port)]

RJ45 port connected via Ethernet cable to the ISP modem or fiber ONT. It receives the public IP address and incoming internet packets.

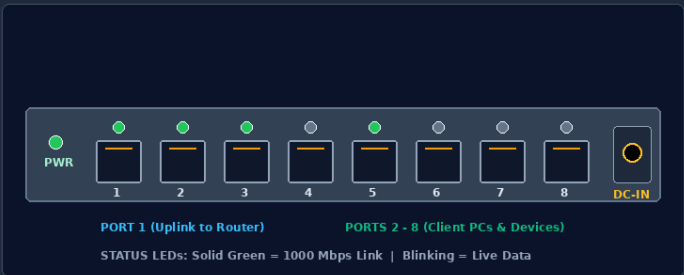
3. Ethernet Port #2: LAN Ports (1-4) [Position: Rear Panel (Yellow Ports)]

High-speed RJ45 Ethernet ports used to connect local wired devices (such as a gaming PC, network switch, or Smart TV) to the private LAN.

4. LED Status Indicators [Position: Front / Top Face]

Visual diagnostics strip showing device state: Power LED (solid green = active), Internet LED (green = online, red = disconnected), and LAN LEDs (blinking = live data).

2. NETWORK SWITCH (8-Port Gigabit Switch)



1. Power Port (DC-IN) [Position: Rear / Side Panel]

2. Ethernet Port #1: Uplink Port [Position: Front Panel (Port 1)]

3. Ethernet Port #2: Client Ports (2-8) [Position: Front Panel (Ports 2-8)]

4. Port Link / Activity LEDs [Position: Directly Above Each RJ45 Port]

Diagram illustrating the identification of Power Ports, Ethernet Ports, and LED Indicators on a Router and Switch. The Router section shows the Power Port (DC-IN), WAN (Internet) Port, LAN Ports (1-4), and LED Status Indicators. The Switch section shows the Power Port (DC-IN), Uplink Port, Client Ports (2-8), and Port Link / Activity LEDs. The status note for the switch indicates: Solid Green = 1000 Mbps Link | Blinking = Live Data.

Task 4: Gaming Café Setup Scenario (8 PCs for IPL Streaming & Multiplayer)

- **Which Device Connects First?**

- Connect the **Router** first to the ISP line (Modem/ONT →Router's WAN port).
- *Reason:* The router is required at the edge to negotiate the public IP with the ISP, manage bandwidth, run NAT, and provide firewall security and DHCP addressing.

- **How to Connect the 8 Gaming PCs:**

1. Run a single Cat6 Ethernet cable from one of the router's **LAN ports** to port 1 (Uplink) of an **8-port or 16-port Gigabit Switch**.
2. Connect all **8 gaming PCs** directly into the remaining ports of the Gigabit Switch using dedicated Cat6 Ethernet cables.

- **Justification:**

- Standard consumer routers only offer 4 LAN ports, which is insufficient for 8 PCs; an external switch expands port capacity without performance loss.
- Wired Gigabit switch connections ensure high-speed throughput (1 Gbps) and ultra-low latency (<10 ms jitter) essential for lag-free multiplayer gaming and buffer-free HD/4K IPL live streaming.